Problem Set 4

Section 3.5

2) Let $V = \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 1 & 1 & 0 \\ 0 & -1 & 0 & -1 & 0 & 1 \\ 0 & 0 & -1 & 0 & -1 & -1 \end{bmatrix}$

$\downarrow$

$\begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & -1 & 0 & -1 & 0 & 1 \\ 0 & 0 & -1 & 0 & -1 & -1 \end{bmatrix}$

$\downarrow$

$\begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & -1 & 0 & -1 & -1 \end{bmatrix}$

$\downarrow$

$\begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

Doing row reduction shows there are only three pivot columns; the first three, which corespond to v_1, v_2 and v_3, thus the max number of independent vectors is three and they are v_1, v_2 and v_3.

20) Our system in the form $Ax = 0$ is:-

$$\begin{bmatrix} 1 & -2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Solving the equation for $y = 1$ and $z = 0$ we get:-

$$x - 2(1) + 3(0) = 0$$
$$x = 2$$

Thus we have $[2 \quad 1 \quad 0]^T$

Solving the equation for $y = 0$ and $z = 1$ we get:-

$$x - 2(0) + 3(1) = 0$$
$$x = -3$$

Thus we have $[-3 \quad 0 \quad 1]^T$

Hence a basis for the given equation would be:- $\left\{ \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix} \right\}$

Because these vectors span the nullspace.

The intersection with the xy plane occurs when $z = 0$, and it is a line. Since $[2 \quad 1 \quad 0]^T$ is in the xy plane, and is a vector in the plane formed by the equation, and it is a line, it is hence also a basis for the xy plane.

~~To find a basis~~ Since the plane is the nullspace of the matrix $A = [1 \quad -2 \quad 3]$ and since the nullspace and row space are orthogonal complements, $[1 \quad -2 \quad 3]^T$ is a basis for the plane (since it is the only row in A).

37) $$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & a & b \\ 0 & d & e \\ 0 & g & h \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = \begin{bmatrix} d & e & f \\ g & h & i \\ 0 & 0 & 0 \end{bmatrix}$$

if the RHS of both equations are equal then we have:-

$$d = g = h = 0 \qquad e = \& i = a \qquad f = b$$

Substituting these values in A we get:-

$$A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = \begin{bmatrix} a & b & c \\ 0 & a & b \\ 0 & 0 & a \end{bmatrix}$$

First blank:- Three (Since only three entries can be independently chosen)

41) $I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$+ \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} * \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} * \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$

Suppose a combination $c_1P_1 + c_2P_2 + \dots + c_5P_5 = 0$ matrix, where $P_1 \dots P_5$ are number in the same order they appear above. Then the left hand side of the equation evaluates to:-

$$\begin{bmatrix} c_1 & c_2 + c_4 & c_3 + c_5 \\ c_2 + c_5 & c_3 & c_1 + c_4 \\ c_3 + c_4 & c_1 + c_5 & c_2 \end{bmatrix}$$

Since this matrix equals the zero matrix, $c_1 = c_3 = c_2 = 0$. We also have the equations:-

$c_2 + c_4 = 0 \Rightarrow c_4 = 0$
$c_3 + c_5 = 0 \Rightarrow c_5 = 0$

Since we have shown that our supposition forces all the scalar multipliers $c_1 \dots c_5$ to be 0, we have proven that $P_1 \dots P_5$ are linearly independent.

44) Dimension of output = r
" " input s = n
" " nullspace = n-r

Section 3.6

11) a) $r < m$ and $r \leq n$

b) If $\dim(N(A^T)) > 0$ then solutions other than $y = 0$ exist. The condition is true when $\dim(C(A)) < m$ i.e. when $r < m$ because $\dim(C(A)) + \dim(N(A^T)) = m$

24) It is solvable when d is in the ~~left nullspace~~ row space of A. The solution y is unique when the left nullspace contains only the zero vector.

28) Rank (B) = 2 and Rank (C) = 2 because row elimination gives 0 rows for all rows except the top 2 in both matrices.

Bases for the row space for both B and C are ~~its~~ their first two rows.

basis for the its
The ↑ left nullspace of B are ~~its~~ special solutions, which are:-

$$\begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} -1 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ -1 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ -1 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} -1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ -1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

The basis for the left nullspace of ~~B~~ are ~~also its~~ the special solutions of C^T (since C is assymmetrical) which are ~~(if rows 1 or~~

$$\begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ -1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} -1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

Similarly, the special solutions of $Cx = 0$ are the basis for the nullspace of C, and those special solutions are:-

$$\begin{bmatrix} -1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ -1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \text{ and } \begin{bmatrix} 0 \\ 0 \\ -1 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

To get the remaining special solutions we have to solve some simultaneous equations:-

$$x_1 r + x_2 n + b = 0$$
$$x_1 + x_2 + 1 = 0$$

which is solved by $x_1 = \frac{b-n}{n-r}$ and

$$x_2 = \frac{r-b}{n-r}$$

Now we solve:-

$$x_1 r + x_2 n + q = 0$$
$$x_1 + x_2 + 1 = 0$$

which is solved by $x_1 = \frac{q-n}{n-r}$ and

$$x_2 = \frac{r-q}{n-r}$$

Lastly we solve:-

$$x_1 r + x_2 n + k = 0$$
$$x_1 + x_2 + x_3 = 0$$

Which is solved by $x_1 = \frac{k-n}{n-r}$ and

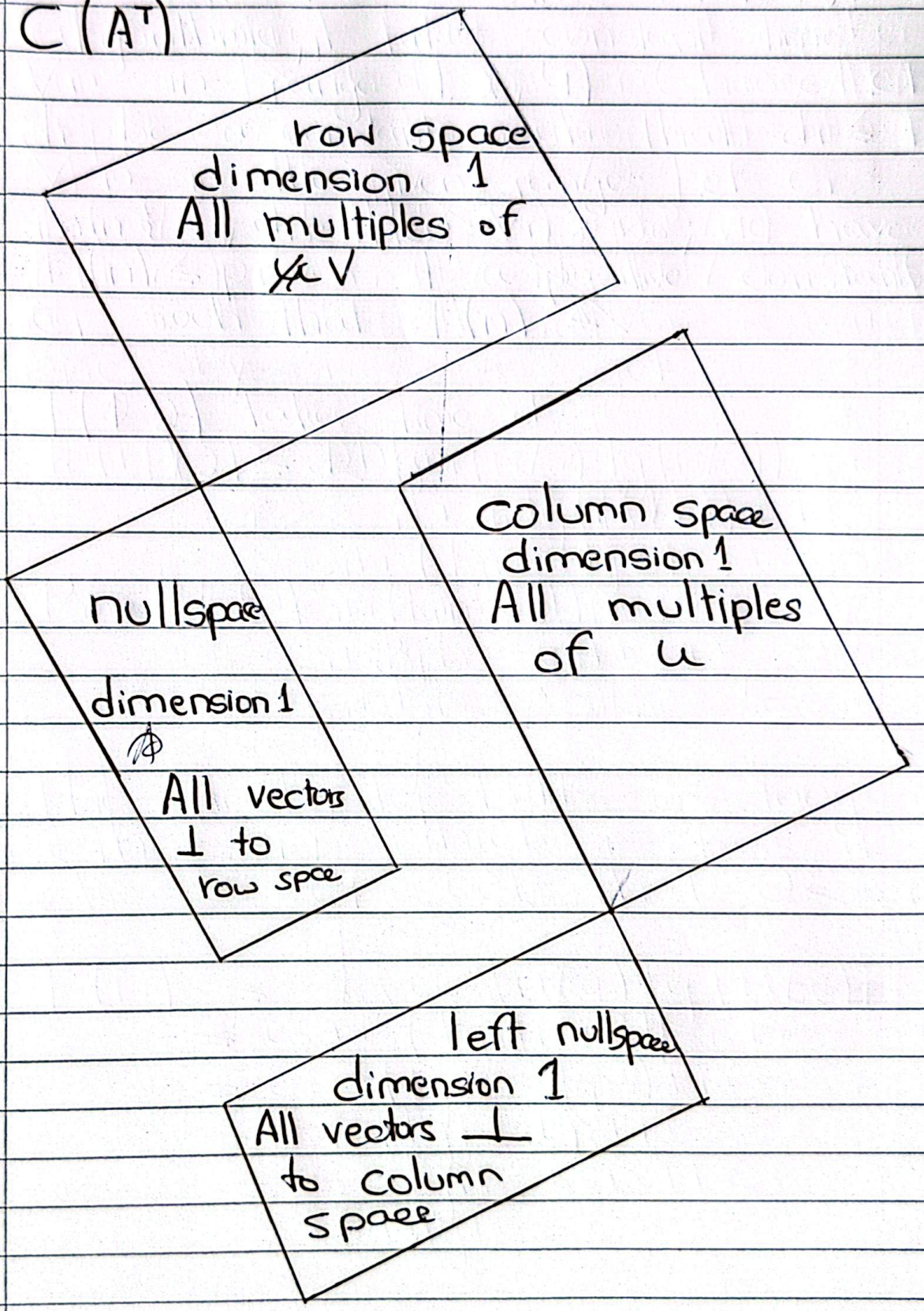
$$x_2 = \frac{r-k}{n-r}$$

So our last three special solutions, which together with the earlier special solutions provide a complete basis, are:-

$$\begin{bmatrix} b-n/n-r \\ r-b/n-r \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} q-n/n-r \\ r-q/n-r \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \text{ and } \begin{bmatrix} k-n/n-r \\ r-k/n-r \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

30)

$C(A^T)$



If B hasse these same four subspaces then $B = u'v'^T$ where u' and v' are some multiple of u and v.

31) a) The matrices where every column is a multiple of $[1 \quad 1 \quad 1]^T$.

b) Matrices where every column is a linear combination of the columns of A.

a) All matrices in the nullspace, as shown earlier, are of the form:-

$$\begin{bmatrix} a & b & c \\ a & b & c \\ a & b & c \end{bmatrix}$$

independently

Only three entries can be chosen independently, so the nullspace has dimension 3.

For the column space, every matrix in it must have columns whose entries sum to 0. Since each column has 3 entries, that means we can independently choose 2 entries in each column, giving a total of 6 independent entries in the matrix, therefore the column space has dimension 6.

They add up to 9 because 9 is the dimension of the 'input space'